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METHOD, DEVICE, AND NETWORK FOR STOPPING BLOOD LOSS

Abstract

A networking-capable emergency tourniquet including an adjustable and inflatable cuff configured to encircle a limb of a patient, and a controller unit removably coupled to the inflatable cuff. The controller unit includes a microcontroller configured to control the introduction, removal, and maintenance of air within the inflatable cuff and process collected patient data, a communicator unit configured to communicate with the user, a transmitter configured to transmit the collected patient data to a centralized server over a communication network, and a receiver configured to receive information from at least one of the centralized server and other sources over the communication network.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/415,106, filed on Jan. 17, 2024, which is a divisional of U.S. patent application Ser. No. 18/368,352, filed on Sep. 14, 2023, which issued as U.S. Pat. No. 12,070,229 on Aug. 27, 2024, both of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to tourniquets and more specifically to a system and method for rapidly treating patients following traumatic injury by use of at least one automated pneumatic emergency tourniquet, where the tourniquet is configured to transmit data to a network.

BACKGROUND OF THE INVENTION

[0003] Rapid access to tourniquets is critical to stop extremity bleeding and prevent morbidity and mortality. Depending on the location of the bleed and severity of the injury, hemorrhage can become life-threatening in seconds. Windlass tourniquets are conducive to neither speed nor accuracy, which are critical to save a life in emergency scenarios. Studies indicate that slow response time, improper placement, and inadequate tightness are significant factors contributing to the failure of windlass tourniquets in controlling bleeding.

[0004] The increase in survival rate made possible by early hemorrhage control is enhanced when paramedics and responders can arrive on the scene quickly. There are over 2 million people trained to stop extremity hemorrhage, but if they are not aware of a nearby extremity hemorrhage occurring they are unable to be of assistance. Thus the most effective way to control hemorrhage is to dispatch the closest tourniquet to the bleed site and to effectively apply it to the limb. Prior art systems describe locators for Automatic External Defibrillators (AEDs), however, AEDs are helpful for treating cardiac arrest, not exsanguinating hemorrhage. Other systems describe location systems for emergency devices such as fire extinguishers, however this is helpful for extinguishing a fire, not for stopping extremity bleeding. Thus a need exists to deliver rapid care to stop emergency bleeding.

[0005] Inadequate tightness is a limitation of windlass tourniquets. Pneumatic electronic tourniquets of the prior art overcome some of the problems with windlass tourniquets, however they are nearly exclusively intended for surgical settings. Prior art systems describe automated pneumatic tourniquets for surgical purposes that provide real-time pressure data and can be adjusted, however they are usually heavy and are not practical for portability. Additionally, the inflatable cuffs designed to be compatible with these systems can cost hundreds of dollars per unit, and the inflation units can cost thousands of dollars. For emergency scenarios, where tourniquets likely become contaminated with bodily fluids or biohazardous materials, it is pivotal that cuffs are disposable and systems are economically feasible, which is a further limiting factor in prior art systems. In other prior art systems, inflation mechanisms and cuffs are described as integral to one another, or have underdeveloped attachment and detachment mechanisms for rapid setup, deployment, and deconstruction. This is not conducive to the speed and access necessary for emergency scenarios. The proper placement and inflation of the tourniquet requires medical credentials and extensive training, making it challenging for non-specialists or first responders to utilize effectively. Additionally, maintaining the necessary air pressure and monitoring the tourniquet's duration during extended field operations can be cumbersome and impractical. Moreover, the risk of potential complications, such as nerve or tissue damage, increases when skilled medical supervision and precise control are lacking. Considering these factors, pneumatic tourniquets have been limited in their ability to be deployed for emergency use.

[0006] Other documents describe a limb occlusion device that has a dual-cuff system. One cuff is to be held at a high enough pressure to stop the flow of blood through the limb, while the other is to recognize the presence of an arterial blood flow through oscillations. Although a mentioned application is toward emergency use in the field, consideration is not given to limitations of using the device in the field: patient movement is virtually guaranteed following the onset of a tourniquet application which makes accurate oscillatory blood pressure measurement highly impractical, therefore the basis for tightening above a limb occlusion pressure is erroneous. A single battery may not be adequate to power the device for an entire application cycle, especially for prolonged care scenarios where application time can exceed 72 hours, and consideration is not given to the type of battery which may be impacted by extreme temperatures.

[0007] Improper placement presents an additional limitation when it comes to tourniquet usage. Research demonstrates that tourniquets should not be positioned over joints and are unsuitable for limbs with multiple bones, such as the forearm. In high-stress situations, it is critical to remind users, particularly those with minimal or no training, about these guidelines. Traditional windlass tourniquets and automated tourniquets from previous designs often rely on physical instruction manuals or device labeling, which are inconvenient to carry with the device and unlikely to be consulted during a rapidly evolving emergency. Consequently, the absence of clear instructions contributes to higher failure rates when applying conventional tourniquets.

[0008] Some prior art documents suggest applications for implementing pneumatic tourniquets into the field without consideration to how this would practically work. For example, the need for tourniquet availability in public settings is clear due to shootings and mass casualty events in locations such as schools, stadiums, offices, and stores, however prior art systems do not describe an effective storage solution for battery powered or rechargeable devices. Additionally, electronic use is described, but consideration is not given to how tourniquet systems will communicate with responders, physicians or other medical personnel. Furthermore, documents fail to describe a practical monitoring solution for a plurality of automatic tourniquets, leaving readers under the assumption that each automatic tourniquet would require manual battery or systems checks, which are time consuming and impractical.

[0009] Thus a need exists to overcome the aforementioned problems with the prior art.

SUMMARY OF THE INVENTION

[0010] The present disclosure provides a system and method for rapidly treating patients following traumatic injury. In one embodiment of the present disclosure, a networking-capable emergency tourniquet is provided. The tourniquet includes an adjustable and inflatable cuff configured to encircle a limb of a patient, and a controller unit removably coupled to the inflatable cuff. The controller unit includes a microcontroller configured to control the introduction, removal, and maintenance of air within the inflatable cuff and interpret patient data, a communicator unit configured to communicate with a user, a transmitter configured to transmit the collected patient data to a centralized server over a communication network, and a receiver configured to receive information from at least one of the centralized server and other sources over the communication network. In some embodiments, the information from the centralized server includes at least one of visual and audible instructions to a user of the tourniquet, which may be delivered through at least one of a speaker or visual display. The centralized server may be in communication with a plurality of network-capable emergency tourniquets. In some embodiments this tourniquet weighs less than eighteen pounds and is therefore ideal for portability. The communicator unit may transmit collected patient data upon activation of the controller, such as in one embodiment the location of the tourniquet and the patient. Patient data may also include elapsed application time, patient blood pressure, tourniquet pressure, or other metrics that may be helpful to treatment. In some embodiments a request for assistance is communicated to an emergency dispatcher through the communicator. The tourniquet may include an RFID tag, which may be configured to store information specific for each tourniquet.

[0011] In one embodiment, the network-capable emergency tourniquets may be stored in a tourniquet storage station. The tourniquet storage station comprises in one embodiment at least one network-capable emergency tourniquet, a station controller configured to aggregate data received from the tourniquet or multiple tourniquets, an internet access point allowing the aggregated data to be accessed via the internet, a transmitter configured to transmit the aggregated data to a centralized server. In some embodiments the station may include a locator configured to report the location of the tourniquet station to the centralized server or to a user. In some embodiments the storage station includes at least one charger to charge each of the network-capable emergency tourniquets. The charger may include an RFID tag that allows an external device to receive and display status and troubleshoot connectivity issues. The storage station may also include mechanical inflators.

[0012] The network-capable emergency tourniquet may be utilized to aid a patient. In one method of use, the tourniquet is programmed with a location. Upon activation of the tourniquet, the network receives the status of the tourniquet, and transmits the location of the patient to at least one of a remote emergency dispatcher and a first responder located within a predetermined distance of the location of the patient. If the tourniquet is moved, the location of the patient is updated, and at least one of the remote emergency dispatchers and first responders is given the updated location of the patient. In one embodiment, the method comprises at least one of verbal instructions and visual instructions being delivered. The location information can be further specified to include the floor and room number indicating where the tourniquet is stored.

[0013] In one method of utilizing the tourniquet, a centralized server is notified of the patient emergency bleeding situation. The centralized server then identifies at least one tourniquet located in the proximity of the location of the patient emergency bleeding situation, wherein the at least one tourniquet is programmed with location information. A request for assistance is then delivered by the centralized server for the location of the patient emergency bleeding situation. Instructions may be delivered through a speaker to a user explaining how to deploy and operate the tourniquet. Further, the centralized server may deliver a notification to an emergency service to be deployed to the location of the emergency bleeding situation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The present disclosure will now be described, for exemplary purposes, in more detail by way of embodiment(s) and with reference to the enclosed drawings, in which:

[0015] FIG. 1A shows an assembled tourniquet;

[0016] FIG. 1B shows the tourniquet separated into two components: the TQController and the TQCuff;

[0017] FIG. 1C shows the inside of a TQCuff;

[0018] FIG. 1D shows a sectional view of the TQController interacting with the TQCuff prior to attachment;

[0019] FIG. 1E shows a sectional view of the TQController interacting with the TQCuff after attachment;

[0020] FIG. 1F shows a TQCuff inflated mechanically without the TQController;

[0021] FIG. 1G shows an alternate TQCuff storage configuration and an embodiment where the TQController includes an RFID reader and tag;

[0022] FIG. 1H shows an alternate TQCuff storage configuration and an embodiment where the TQController includes an LCD screen;

[0023] FIG. 1I shows an alternative tourniquet embodiment where the valve is located at a 90 degree angle and the shape of the controller is square;

[0024] FIG. 1J shows a sectional view of attachment for the embodiment demonstrated in FIG. 1I;
[0025] FIG. 2A shows a circuit diagram of the TQController;
[0026] FIG. 2B shows the electronics stored in the TQController;
[0027] FIG. 2C shows the electronics stored in an alternative embodiment;
[0028] FIG. 2D shows pictorial instructions placed on top of the tourniquet in some embodiments;
[0029] FIG. 2E shows a method in which the tourniquet is applied after bleeding is identified by a witness;
[0030] FIG. 3A shows a tourniquet station where full tourniquets are stored;
[0031] FIG. 3B shows a TQStation where TQCuffs and TQControllers are stored separately;
[0032] FIG. 3C shows exemplary TQStation circuitry;
[0033] FIG. 3D shows an exemplary TQStation display;
[0034] FIG. 4A shows a user application;
[0035] FIG. 4B shows another screen of a user application;
[0036] FIG. 4C shows a Responder application;
[0037] FIG. 4D shows a phone notification requesting tourniquet dispatch;
[0038] FIG. 4E shows a phone application;
[0039] FIG. 5 shows the TQNetwork; and
[0040] FIG. 6 shows multiple methods in which emergency bleeding is stopped with the tourniquet.

DETAILED DESCRIPTION

[0041] In the following, in several scenarios, a detailed description is presented of the exemplary embodiments in conjunction with the accompanying drawings to enable easier understanding of the solution(s) described herein.

[0042] FIG. 1A provides a detailed view of the pneumatic tourniquet **100**, which consists of two components shown separately in FIG. 1B: the TQController (“or “controller”) **102** and the TQCuff (or “cuff”) **104**. The TQController **102** adds air into the TQCuff **104**, which pressurizes the cuff when secured about a limb of a patient allowing it to slow and stop the bleeding.

[0043] The TQController **102** comprises a power switch **126** to power the tourniquet **100** on and off and a charging port **129**. The power switch can be alternatively programmed to toggle between idle and active mode, rather than on and off mode in one embodiment. An LED indicator light **125** may be used to communicate readiness/non-readiness, battery life, etc., via a visual indication, such as, for example, color. Exemplary pictorial directions **209** are adhered to the top of the TQController **102** in some embodiments. In one non-limiting embodiment, they contain four steps to be completed in succession: [0044] 1. Apply direct pressure to bleed. [0045] 2. Tighten high above bleeding parallel to the ground. [0046] 3. Power on and inflate “+” [0047] 4. Get medical help!

[0048] In this embodiment, a series of controls, buttons, or switches are located on one or more sides of the TQController **102**. There are two buttons in this embodiment. The “inflate” button **130** with corresponding label **205** controls the addition of air into the cuff **104** when prompted, which stops the bleed by applying pressure to compress the artery and the vein. In one embodiment, the tourniquet **100** includes a speaker that delivers audible instructions, which can be silenced by selecting the “mute” button **128** with corresponding label **203**. It may be desirable to mute the audible instructions in some applications, such as, for example, in an active shooter situation or a military application. These audible instructions are explained in further detail below and with regard to FIG. 2E.

[0049] The controller **102** may comprise multiple buttons or controls with similar or different labeling, and the placement of controls may vary. Other examples of controls include, but are not limited to, those for contacting a dispatcher, decreasing pressure, or detaching the cuff **104** from the controller **102**. Each of these can use universal symbols above or directly on the button. In certain embodiments, the control may comprise elements such as a depressible button, conductive button, switch, touch-sensitive interface, dial, or any combination thereof, configured to provide response

input or command signals to the system. Alternatively, some or all controls may be removed in exchange for controls in a mobile application such as in FIG. 4E.

[0050] FIG. 1B shows the TQController **102** separated from the TQCuff **104**. To secure the controller **102** onto the cuff **104**, the cuff's integrated slider **108** is moved vertically into the TQController's receiver **110** in the direction of arrow **115**. The components are securely positioned by bringing the cuff **104** and controller **102** towards each other until top valve **114** is securely in the TQController **102** following arrow **113**. This integrates the top valve **114** with the TQController **102**, allowing fluid access between the TQCuff's inflatable member **112** (shown in FIG. 1C) and the TQController's air pump **219** shown in FIGS. 2A-2B. The tourniquet **100** is fastened onto a limb by advancing the distal end of the TQCuff **104** through a loop **117**, and securing it to the side of the cuff **104** which in some embodiments is lined with Hook and Loop **121**. When directed, the TQController **102** then adds air into the TQCuff **104** to pressurize the tourniquet **100** and stop the bleed via arterial and venous compression. Cuff fabric **119** encases the inflatable member **112** embedded into the cuff **104** and is shown in more detail in FIG. 1C.

[0051] In certain embodiments, TQCuff **104** can be disposed of after each use to stop a bleed and replaced with a new cuff. This reduces the likelihood of pathogen transmission between patients. The TQCuff **104** can be manufactured in various widths and lengths, and tailored to accommodate limbs of different sizes. For instance, a narrower and shorter size is suitable for pediatric limbs, while a broader and longer size is better suited for adult limbs. These specifications can also be adjusted based on the specific limb type, such as a size customized for a pediatric patient's arm or an obese patient's leg. A Radio Frequency Identification (RFID) tag or similar identification means may be embedded into the cuff to store various cuff information, including the length, width, suitable inflation pressure, model number, lot number, serial number, production identifier, etc. Inclusion of an RFID tag would allow an RFID Reader incorporated into the TQController **102** to interpret the stored data and inflate the tourniquet faster and more effectively.

[0052] An expanded view of the components of TQCuff **104** is shown in FIG. 1C. To achieve arterial and venous compression, the TQCuff **104** incorporates a limb conforming component **106** to facilitate rapid attachment to a limb. The limb conforming component **106** includes an integrated dual port air valve, comprising a top valve **114** and a bottom valve **116**. In this embodiment, there is one bottom valve **116**, but this can be replaced in favor of multiple bottom valves with multiple bladders as a fail-safe in case one bladder breaks. In some embodiments the valves can be one-way valves to prevent air leakage. Integrating both the top valve **114** and bottom valve **116** directly into the limb conforming component **106** minimizes the need for external components, reduces manufacturing time, and reduces the likelihood of breakage or faulty connections between components.

[0053] In one embodiment, TQController **102** housing and the limb conforming component **106** can be fabricated from stereolithography (SLA) printing with photopolymer resin. This manufacturing process ensures the creation of seamless and durable components. If resin is printed using SLA, the air valve hole may be cleared by first rinsing the print in Isopropyl Alcohol (IPA) as is standard procedure, then introducing air through the valve to clear out additional resin or IPA. Varying curing times with ultraviolet (UV) light to increase polymerization of the material post production allows for a higher degree of toughness and thus a longer product lifetime.

Alternatively, these parts can also be fabricated through techniques such as injection molding, vacuum forming, powder printing, fused deposition modeling (FDM) 3D printing, or other suitable methods. The controller housing is constructed from a resilient material, allowing it to withstand repeated use. It is designed to be easily cleaned and sanitized for multiple applications. An exemplary cleaning method for the TQController **102** involves utilizing a bleach and water solution that is commonly employed in the sanitization and maintenance of reusable medical devices.

[0054] The inflatable member **112** is secured beneath the limb conforming component **106**. The inflatable member's tubing **118** extends above the inflatable member **112** and is slid over the bottom

valve **116**.

[0055] FIG. **1D** shows a cross sectional view of the TQCuff **104** and TQController **102** before attachment. During the manufacture of TQController **102**, tubing is fastened inside guide **122**, prior to use. The connection between tubing **118** and tubing stored in tubing guide **122** and the valves **114** and **116** is reliable due to the shape of the valves **114** and **116**. An airtight seal is created because multiple inverted triangles of which the bases are slightly larger than tubing are integrated into the valves. This connection may be reinforced with an adhesive, silicone, cured resin, small zip tie, or other suitable fastener. A printed circuit board (PCB) **215** contains a microcontroller **210** that is responsible for controlling the circuitry **200**, shown in detail in FIG. **2A**. In certain embodiments, the microcontroller **210** is an ESP32 chip, possessing features such as wireless internet connectivity, Analog-to-Digital conversion (ADC), multiple sleep modes for reduced power consumption, and secure communication capabilities for establishing encrypted connections such as Transport Layer Security (TLS) connections. When connected to the internet, the device can utilize Voice Over Internet Protocol (VOIP) or real-time communication protocols to establish direct connections with emergency services. The microcontroller **210** facilitates data collection and records the time of tourniquet application, pressure levels, and other critical data.

[0056] The inflatable member **112** achieves connection with the controller **102** by the tubing **118** being fastened over valve **116**. To set up a tourniquet **100** for use, connection between the TQCuff **104** and TQController **102** is mechanically achieved. In FIG. **1D**, it can be seen that the slider **108** is integral to the limb conforming component **106**. The slider **108** and receiver **110** guide the top valve **114** to enter the tubing **118** located within the tubing guide **122** when attached as indicated by arrow **115**.

[0057] FIG. **1E** shows the result of the TQController **102** and TQCuff **104** being integrated from a cross section view. When selected, the “inflate” button **130** shown in FIG. **1A** sends an “inflate” command through the microcontroller **210**. The microcontroller **210** then directs the air pump **219** shown in FIGS. **2A-2B** to release air to pressurize the inflatable member **112**, which firstly travels through the air pathway **226**, then through tubing guide **122**, then bottom valve **116** and tubing **118**, and finally causes the inflatable member **112** to inflate, which pressurizes the tourniquet cuff and reduces the flow of blood through the limb. Thus when the TQController **102** and the TQCuff **104** are attached, a fluid connection is formed that allows for stopping a bleed.

[0058] In some embodiments, the top valve **114** and tubing guide **122** are equipped with an attachment sensor, such as a colorimetric, capacitance, mechanical, or electrical touch sensor. This can provide an audible sound to indicate proper attachment during setup, or a visual indicator of proper attachment, such as a light change or other color change. The cuff **104** may further include a mechanical indicator when this attachment is achieved, indicating that the cuff **104** has been used and should not be re-used, such as a mechanical latch released to change a color indicator.

Alternatively, the tourniquet **100** may conduct a pressure test by inflating the cuff's inflatable member **112** to determine if target pressure is achieved without the presence of a limb. If successfully inflated and pressure is maintained, it would follow that the connection is reliable. If the inflation test proves to be unsuccessful, an alert will be delivered from the centralized server **500** in FIG. **5** that the connection is unreliable. If this failure persists, further support will be necessary. This inflation test can be repeated at preset intervals, such as daily, to ensure that the components are ready for use. It is encouraged that attachment between the controller **102** and cuff **104** is secured prior to using the tourniquet **100** to facilitate speed when application is desired.

[0059] After using the tourniquet **100**, the controller **102** and cuff **104** may be disengaged by guiding the slider **108** away from the receiver **110** by retracing the action demonstrated by arrow **115**. This effectively terminates the connection between the cuff **104** and controller **102**, facilitating the disposal of the cuff **104** and connection of a new cuff **104**. Disposing the cuff **104** after use decreases the risk of infection or pathogen transmission between patients.

[0060] The dual valve system comprised of top valve **114** and bottom valve **116** may be designed to

maintain pressure independent of the controller **102**, so that the controller **102** can be used to inflate multiple cuffs **104** consecutively without needing constant attachment. This can be achieved by adding a check valve or one way direction to the integrated valves. This would allow the cuff **104** to operate independently of the controller **102**, keeping pressure without constant regulation. This would be especially useful in a mass casualty incident where there are potentially a plurality of victims and a shortage of controllers **102**. Additionally, this may save power during attachment, especially in the case of leakage from an electronic component where the system may compensate for air loss. Cuff fabric **119** is sewn around both the inflatable member **112** and the limb conforming component **106**, preventing the inflatable member **112** and limb conforming **106** component from being detached during use.

[0061] In another embodiment shown in FIG. **1F**, a mechanical bulb **133** may be attached to the top valve **114** on the limb conforming component **106** rather than the controller **102** for inflation. The bulb **133** must be repeatedly squeezed to add air to the inflatable member **112**. This may be desirable for a military application where displacement of casualties makes charging or electronics difficult, or as a redundancy in the case of electronic failure. The bulb may be integrated with the electronics so pressure can still be monitored during use.

[0062] FIGS. **1G** and **1H** show alternative embodiments of the tourniquet **100**. The cuff **104** may be stored in a “quick launch” configuration where it is secured through loop **117** and ready to be applied to a limb without full detachment. In some embodiments, both the TQCuff **104** and TQController **102** may incorporate RFID tags **224** or RFID readers **225**. Here, the TQController **102** is shown having an RFID reader **225**, and the TQCuff **104** is shown having an RFID tag **224**. In other embodiments, TQController **102** includes RFID tag **224** and TQCuff **104** includes RFID reader **225**. The RFID tags **224** securely hold fixed data such as, but not limited to, cuff length, ideal pressure, serial number, or others, ensuring accurate identification and tracking of the tourniquet. RFID tag **224** may be read by various devices, including but not limited to cell phones, controllers, TQStations such as in FIG. **3**, or other compatible mechanisms. This fixed information is further accessible through the App shown in FIG. **4**, or may be printed onto the device by a sticker or another label. RFID tag **224** or RFID tag reader **225** may be used to identify the tourniquet **100** easily within a TQStation shown in FIG. **3**, for example, by scanning it upon entry.

[0063] FIG. **1H** depicts the TQCuff **104** in a compact storage state. The TQController **102** incorporates an LCD Screen **131**. LCD screen **131** can facilitate data presentation, encompassing time duration, exerted pressure, and comparative analysis between target and actual pressures to identify variances. In subsequent figures, the LCD screen **131** has capabilities to interface with emergency dispatch systems. Prior to operational use, LCD screen **131** can display battery status, software updates, version details, and other pertinent device information. Moreover, LCD screen **131** can serve a training function, offering live feedback on application precision during utilization.

[0064] FIG. **1I** demonstrates another embodiment of the tourniquet **100**. Control **132** and corresponding label **207** are for deflation of the tourniquet **100**, which is particularly desirable for a medical professional where tourniquets are often removed to assess bleeding and then reapplied if necessary. The placement of upper valve **114** is at a 90 degree angle in this embodiment, and the sliding mechanism **110** begins on the edge of the TQController **102** rather than in the center. The slider **108** slides directly into the side of the case in this embodiment following arrow **115**. Those skilled in the art will recognize that there are a variety of placement alternatives for where the cuff **104** may enter or where upper valve **114** may be configured on cuff **104** and the end goal of fluid integration is preserved.

[0065] Additionally, the “deflate” button **132**, “inflate” button **130**, “mute” button **128** and “power” button **126** controls may be switched for alternative controls. In the case that the TQController **102** does not have an LCD screen **131** as in FIG. **1H**, it may be desirable to include a control that would allow the TQController **102** to communicate status through a speaker, such as time applied, pressure applied, etc. This embodiment includes a speaker **142**, which is optional depending on the

material with which the TQController **102** is manufactured. For example, it may not be necessary to include speaker holes if the material is thin enough to hear through. When application is desired, pictorial instructions **209** are provided to facilitate rapid application and speaker **142** delivers further instruction. The instructions may alternatively be communicated through an external device such as a nearby phone, or may be limited to the pictorial instructions affixed to the top of the case. This unit may also contain a charging port **129**.

[0066] FIG. **1J** demonstrates a sectional view of the tourniquet embodiment shown in FIG. **1I**. The TQCuff's slider **108** integrated into the limb conforming component **106** enters into the TQController's receiver **110**. The upper valve **114** enters into the tubing guide **122** along arrow **115**, where the air pump **219** shown in FIGS. **2A-2B** then pressurizes the inflatable member **112** under direction from the microcontroller **210** shown in FIGS. **2A-2B**. To remove this connection, the units are disconnected following arrow **115**. Like FIGS. **1D** and **1E**, the inflatable member's upper tube **118** is fastened around the limb conforming member's internal lower valve **116**. The electronics **200** then controls the inflation process.

[0067] FIG. **2A** illustrates a block diagram showing the electronics contained inside of TQController **102**. To facilitate the inflation, deflation, and pressure regulation of the cuff's inflatable member **112**, the electronics **200** incorporates an air pump **219**, a pressure sensor **220**, and a valve **222**. These components are connected to the PCB **215** and controlled by the microcontroller **210**. The air pressure introduced into the inflatable member is monitored by a pressure sensor **220**. In certain embodiments, valve **222** is an electronic valve designed to regulate the direction of air flow such as a solenoid. Valve **222** offers flexibility, allowing for manual adjustments or electronic control, thereby enabling precise management of air flow. The pressure sensor **220** indicates when optimal pressurization is achieved, allowing the microcontroller to navigate to the next logic point demonstrated in FIGS. **2E** and **6**.

[0068] The electronics **200** includes power components including a power supply **212**, power regulator **214**, port **129** and power switch **126**. The power supply **212** may be in some embodiments a rechargeable battery such as a lithium-ion polymer (LIPO) battery compatible. Alternatively, the power supply may be a disposable battery, battery pack, or AC/DC power supply. If the tourniquet is used in conjunction with a TQ Station in FIG. **3**, the battery will likely be rechargeable, and it may include a port **129** to receive power. In some embodiments the port **129** is a wireless charger. The power supply **212** is regulated by a power regulator **214** and activated by a power switch **126**. In some configurations, the tourniquet **100** includes an "idle" mode, where it can be on but in a power saving state to preserve battery life, in which case activating any control would awaken the tourniquet from idle mode. Alternatively, once the tourniquet **100** is applied to a limb and the pressure sensor **220** detects a change in pressure, the tourniquet **100** may awaken automatically. The power supply **212** and regulator **214** can incorporate advanced power management features, such as energy harvesting from ambient sources or adaptive voltage regulation systems to optimize power efficiency and extend battery life. These power saving sources and ambient harvesting sources are especially beneficial in prolonged care (PC) scenarios where casualties may be displaced, or active combat theaters where medics are unable to access insecure areas.

[0069] The PCB **215** includes several components. Transmitter **206** transmits collected data from the TQController **102** to a centralized server **500**, shown in FIG. **5**, allowing outside access to data and explained in detail below. Various communication methods can be employed to transmit data from the tourniquet **100** to the centralized server **500**, such as the microcontroller's optional Bluetooth module, as explained previously. Alternatively, the transmitter **206** can be a SIM Card, enabling connection to emergency services such as 911 through a cellular network. Alternatively, long range radio (LORA) or a similar method can be employed. Data can be aggregated into a SQL database or other acceptable data storage configuration.

[0070] Receiver **202** is configured to receive data or communication from an external source, such as data stored in the centralized server **500**. This data can range from hardware and software

version to the patient's ideal limb occlusion pressure (LOP).

[0071] Communicator **135** allows for the delivery of stepwise instructions. In some embodiments the tourniquet allows a 911 dispatcher to communicate with those at the scene of the emergency. This is particularly useful in the case of multiple injuries on one casualty, a mass casualty incident, or if a patient is self-applying a tourniquet **100**. The communicator **135** may be a speaker **142** shown in FIG. **1I**, an LCD screen **131** shown in FIG. **1H**, or an LED light **125** shown in FIG. **1A**. Alternatively, the communicator **135** may be an extraneous device such as a mobile phone or computer application where instructions and status are delivered separately from the tourniquet **100**, or may be disabled in favor of the pictorial instructions affixed to the device **209** in FIG. **1A**.

[0072] Locator **204** can use various methods to provide location data, such as cell tower triangulation, GPS data, Wi-Fi or Bluetooth positioning, or beacon-based positioning. The locator **204** may be removed depending on the microcontroller's locating capabilities. The location can be pre-set upon device configuration and thus live location is not needed. Alternatively, if cellular connection is enabled, the controller **102** can use Automatic Location Identification (ALI), which is used by Public Safety Answering Points (PASPs) to receive and display information about the caller, thus eliminating the need for a separate locator. PASPs are usually monitored by dispatchers who can promptly dispatch responders to the given location. The specific implementation and infrastructure can vary between countries and regions. For example, it may be desirable to have a more rugged GPS based locator for a military application where casualties may be in remote areas, or to eliminate the locator in a suburban or metropolitan area in favor of reliance on ALI or Wifi location. Capability exists for the tourniquet **100** to connect remotely to another GPS location device, for example, to a cell phone's Bluetooth or hotspot, a boat's EPIRB system, or an independent GPS locator's system. If used in conjunction with a TQStation in FIG. **3**, location capabilities may be overridden in favor of pairing with the locator or preset location on the TQStation.

[0073] Location capability may be utilized to find casualties displaced within a building, which is particularly helpful in the case of buildings with multiple floors or rooms. This would be executed by programming the tourniquet **100** with the floor and room number it is stored in; activating the tourniquet **100**; transmitting the location and status of the tourniquet **100** to the network upon activation; providing a location database for the building, wherein the location database comprises information on the floor plan and room numbers of the building; receiving the location and status information of the tourniquet **100** from the transmitter **206**; transmitting the location information of the casualty to at least one remote emergency dispatcher and at least one first responder within the vicinity of the building; updating the location of the casualty in the network if the tourniquet **100** is moved; and providing instructions to the dispatcher and first responder to locate and assist the casualty based on the location information received. Advancements in technology may further change the methods used to communicate.

[0074] Advanced pressure control algorithms can be employed, ensuring precise and automated pressure adjustments based on the specific needs of the patient. This can take into account a patient's blood pressure and calculate personalized pressure, such as limb occlusion pressure (LOP). LOP is common practice for finding the minimum effective tourniquet pressure for a patient in controlled environments, such as operating theaters. Finding LOP in the field is difficult unless the patient is very still. Therefore, blood pressure values may be uploaded to the tourniquet's database during initialization to calculate the necessary pressure more accurately. The tourniquet **100** may incorporate additional sensors to detect undesired blood flow, such as a secondary pressure sensor, a small Doppler ultrasound, etc., or incorporate feedback mechanisms for monitoring and sustaining optimal pressure levels during usage. RFID tag **224** and RFID reader **225** shown in FIG. **1G** may further be incorporated into the electronics **200** to facilitate rapid inflation to an ideal pressure. These may be used to store and read additional tourniquet information as aforementioned.

[0075] Further, in a scenario where casualties do not have rapid access to medical treatment, it may be desirable to initiate a partial or complete reperfusion protocol after a certain amount of time has passed to reinstitute blood flow to the limb. This can occur at preset time intervals or can be done on demand through the mobile application, and in some embodiments can be preset through a button or other feedback mechanism. More basic pressure adjustments are accomplished using the included inflate button **130** and deflate **132** button. These buttons and the mute button **128** can be removed in favor of controls on a mobile application, which may be desirable to avoid a patient accidentally activating an inflate or deflate control, thus loosening or tightening the tourniquet **100** unintentionally. The inflate and deflate features may incorporate safety mechanisms that require additional confirmation to inflate or deflate prior to executing the command.

[0076] FIG. 2B demonstrates an exemplary TQController **102** circuitry **200** storage means. Here, an integrated air pathway **226** accomplishes rapid inflation of the cuff **104**, eliminating the use of multiple independent tubes which can be complicated and time consuming to manufacture. PCB **215** is located across the side of the TQController **102**, facilitating rapid connection between the air pump **219**, valve **222** and battery **212**. Each component is slid into the case and can either be soldered or snapped onto the board, which reduces manufacturing time significantly. Additionally, the case is manufactured to easily slide each component in, which stabilizes each component and reduces the likelihood of breakage.

[0077] FIG. 2C shows an alternative storage embodiment for TQController circuitry **200**, which further comprises controls **128**, **130**, and **132**. This embodiment is curved to mold to the shape of a limb. The air pathway **226** is integrated into the bottom of the case rather than the side of the case. The power switch **126** is integrated into the bottom of the case to prevent accidental shutdown during application. The air pump **219** and valve **222** are placed across from one another and here are snap-fit into the case rather than slid into housing. The PCB **215** includes a port **129**. The battery **212** is placed adjacent to the board before the case is closed.

[0078] FIG. 2D shows the pictorial instructions **209** in two different embodiments; one includes control labels **201**, **203**, **205**, and **207** dispersed around the edges of the controller, while the alternative embodiment includes fewer control labels **203**, **205** and **207** on only one side of the controller. There are a number of different arrangements that could be made out of the pictorial instructions that would still preserve the functionality of the system. It may be desirable to place controls on opposite sides to prevent accidental selection of the wrong control, or to eliminate the controls altogether in favor of using controls in the mobile application shown in FIG. 4E.

[0079] FIG. 2E demonstrates exemplary methodology of how tourniquet **100** can stop blood loss. First, an emergency bleed occurs, at step **228**. The tourniquet **100** is then secured by sliding the cuff fabric **119** through the loop **117**, pulling the fabric as tightly as possible, and securing the hook and loop **121** by doubling the strap back as shown in FIG. 1. The device is powered on or awakened out of idle mode at step **230**. Questions are posed through the communicator **135** shown in FIG. 2A. Alternatively, the TQController **102** may be configured to identify when the tourniquet **100** has been applied since the pressure sensor will detect the increase in pressure from manual tightening, and it may automatically awaken out of idle mode.

[0080] Once the tourniquet is powered on, the communicator instructs the user to “tighten strap above bleeding, then press inflate,” or words to that effect, at step **232**. When the “inflate” command is received at step **234**, the tourniquet tightens to a preset pressure, at step **236**. The preset pressure may be obtained through varying methods, such as a database of pressures corresponding to different cuffs, customized pressure, or an RFID tag associated with the particular cuff. The tourniquet states, at step **238**, “If bleeding continues, press “inflate”. If bleeding has stopped, do not remove the strap. Get medical help immediately,” or words to that effect. If bleeding has not ceased, the “inflate” command is received again at step **240**, the tourniquet adds pressure, and reposes the question **238**. If bleeding has ceased, the device delivers a prompt to seek medical help, at step **242**.

[0081] In some embodiments, instructions may be triggered by sensors that verify proper use of the tourniquet. For instance, if the device is powered on before being properly positioned above the bleeding site, the device may verify correct placement before initiating the tightening process by detecting a base pressure as aforementioned. The controls may also be moved and perform similar functions. For example, the deflate control may be hidden or may be removed in favor of control from an external device to prevent accidental deflation of the tourniquet. The pictorial instructions **209** in FIG. 2D are complemented by instructions conveyed through the communicator **135**. In certain embodiments, the pictorial instructions **209** can be substituted with a touch screen or a screen **131** such as in FIG. 1H, enabling the display of interactive instructions. Those skilled in the art will recognize that there are multiple suitable methods for delivering instructions or formatting them.

[0082] FIG. 3A shows a TQStation **300** in front view and in a back view **308**, where a plurality of tourniquets **100** may be contained. The TQStation **300** contains docks **302** and charging mechanisms **304**. Twelve docks **302** are shown in FIG. 3A, however this is not intended to be limiting as the stations can be built with a varying number of docks **302**. In this non-limiting example, the docks **302** are circular, where the tourniquets **100** can be stored in a pre-looped configuration such as in FIG. 1G or 1I. They can then be secured around the docks **302** and easily pulled off when use is desired. A monitor **306** communicates data, shown in more detail below.

[0083] The charging mechanisms **304** here are wireless chargers, but can alternatively be a USB port or other connector. The back of the TQStation **308** includes interconnected charging circuitry **310** that routes each station **308** to a central charging outlet **312** regulated by the station controller **314**. The central charge may be from a wall mount AC or DC power supply, or a remote or a portable power source.

[0084] FIG. 3B shows an alternative front view **301** and back view **309** embodiment of the TQStation **300**. The configuration of the docks **302** can be in a number of shapes, such as squares, slots, or holes. FIG. 3B demonstrates that the TQStation **300** can be fit solely for controllers **303** and have storage **307** for cuffs of varying sizes. Additionally, an alternative charging mechanism **311** where the tourniquet **100** is plugged in via USB to charge is demonstrated in FIG. 3B. This unit contains equivalent back circuitry **310**, outlet **312** and controller **314**. However, the communicator **329** is no longer a screen in this embodiment but is replaced with a speaker.

[0085] Here, two embodiments of the TQStation **300** have been shown, however these may be adjusted to fit a number of applications. It may be useful to exchange some components in favor of fully remote components that allow for use off the grid. Power units may be replaced with solar charging units in the case of a wilderness application. For a vehicular application, it may be useful to reconfigure the TQStation **300** to fit in a car trunk, or in an ambulance or other emergency response vehicle for rapid use in an emergency situation.

[0086] FIG. 3C is a block diagram showing the components of the station controller **314** in detail. The controller **314** aggregates data received from the tourniquets **100** stored in the TQStation **300**. The controller **314** is managed by a microcontroller **326**. Tourniquet data can be received by firstly connecting to the connector **321**, which is in some embodiments an embedded hotspot. Secondly, data is processed by the microcontroller **326** after connection is established. In the event that a tourniquet **100** is loaded onto the dock **304** without prior initialization or setup, it is optional to synchronize data with the TQ Station **300**. For instance, if a tourniquet **100** is placed within a TQ Station **300** positioned near an entry point in a stadium, the communicator **329** may ask whether the tourniquet **100** should be permanently associated with this location. This communication can be accomplished audibly through a speaker, LED light arrangement, or a monitor **306** in FIGS. 3A and 3D. If confirmed, the fixed location data within the tourniquet **100** will be updated to align with the station's data. Stations **300** have the capability to override previously stored location data when tourniquet **100** is transferred from one station **300** to another.

[0087] A transmitter **320** communicates data about the tourniquets **100**, including the

above-mentioned fixed and variable metrics, to the centralized server **500** in FIG. 5. Data can then be aggregated into a SQL database or other acceptable data storage configuration.

[0088] A locator **318** reports the location of the tourniquet station **300** to the centralized server **500**. The connector **321**, microcontroller **326**, locator **318**, and transmitter **320** can use any of the methods previously described to obtain and report location data and communicate it with the centralized server **500** in FIG. 5.

[0089] The controller **314** is connected to a power supply **324** that can be provided via outlet **312** in FIGS. 3A-3B. A power regulator **328** may be included. Each tourniquet **100** may be in the TQStation **300** in either “off” mode, “on” mode, or “idle” mode. In some embodiments, the front of each tourniquet's charger **304** or **311** or dock **302** or **303** may contain an RFID tag that allows an external device to receive and display status, and troubleshoot connectivity issues through the site. Here an RFID reader **323** can be wired to each port or to one location. Alternatively, a motion sensor or mechanical switch may be present to detect a tourniquet.

[0090] Once a tourniquet **100** is employed in an emergency, the station's controller **314** sends an alert to the centralized server **500** that the tourniquet **100** is in use, and this can simultaneously be reported to each connected device with access to the TQStation **300** via a text message, phone call, or other acceptable alert. The TQ Station controller **314** may automatically update how many tourniquets **100** are available for use to the centralized server **500** based on how many tourniquets **100** are present within the station **300**.

[0091] FIG. 3D shows the monitor **306** in more detail. Here the monitor **306** displays the sync status **334** and battery health **336** of the tourniquets contained in the station **300**. In this example, one tourniquet **100** has failed to sync, and thus an alert **332** is displayed indicating a problem with sync status. The labeling convention of alert **332** follows a standard convention labeling the type of building, floor, classroom, and tourniquet number, but since this system can be used in a variety of settings and building types the labeling can be adjusted to reflect each. Alerts can be shown for a variety of device checks and failures. Here, the primary indicators of health are sync status and battery, however indicators may be programmed to show any number of notifications such as expiration dates of components, last use, updates needed, etc. If a bleed is identified, the “click here to stop a bleed” button **338** can be engaged to activate the centralized server and request emergency medical response (see FIG. 6).

[0092] FIG. 4A demonstrates a tourniquet **100** monitoring embodiment through the User App **400**, which shows the location and health of the tourniquets **100**. FIG. 4A is a non-limiting example of a home view of the User App **400** for a school. This can be reformatted to adjust for the type of organization, for example, an office, gym, stadium, airport, etc.

[0093] A summary of tourniquet statistics is depicted across the top, showing total device readiness **402**, the number of tourniquets deployed **404**, the number of bleeds stopped **405**, and the number of users trained to stop the bleed **407**. These can be reordered to suit the organization's individual needs.

[0094] Alerts **410** include software updates, tourniquet use history, expired components, and other indicators. In this example, the most recent notification shows that a unit failed to connect **411**. This is further reflected in Sync Status **412**, indicating that one unit failed to sync.

[0095] Battery levels **414** are shown for the tourniquets **100**, and a graph of tourniquet readiness **416** indicates the total device readiness over time. Map **406** includes all of the major tourniquet stations **300** aggregated to a building level. Circle **408** shows the building where the tourniquet **100** failed to connect. This may be a different color than the “ready” units, and here is outlined in black.

[0096] FIG. 4B demonstrates what occurs once the highlighted circle **408** or the alert notification **411** in FIG. 4A is selected. The site navigates to screen **418**, showing a detailed view of the building where the disconnected tourniquet **100** is located. Summary box **420** shows the classroom, floor number, and failed tourniquet unit number, and can optionally show other information, such as the nearest emergency exits, or emergency assembly points. In this embodiment, data is overlaid

on map **421**. Alternatively, the site can display a floor plan with the tourniquet's location highlighted or other architectural modeling that for easy navigation through a building. Advantageously, this reduces the need for reliance on an external mapping platform, however if more precise navigation is necessary, "Take Me There!" **422** may open an external map platform. [0097] Those skilled in the art may recognize that there are a number of configurations that will communicate the same essential information about a tourniquet population, and the present embodiment disclosed is not meant to be limiting.

[0098] FIG. **4C** shows the Responder App **424** in detail. The primary difference between the User App **400** and the Responder App **424** is that the Responder App **424** has permissions to view many tourniquets **100** in an area. The Responder App **424** is further configured to quickly facilitate the dispatch of a tourniquet **100**, as time is of the essence to stop blood loss. In the event of a hemorrhage, the "Emergency" section **426** describes an emergency **428** where one tourniquet **100** is required. The "Emergency" section displays the address and coordinates of the emergency. [0099] In one embodiment, the system encompasses optimal tourniquet selections, specifically designated as **430** and **431**, which afford the operator with the premier means for deployment in an emergent situation **436**. Whilst the proximity and transit time to the emergency are of importance, there are a multitude of parameters pivotal in the decision-making process for tourniquet deployment.

[0100] A consideration includes the energy reservoir level of the tourniquet **100**; if found deficient, the tourniquet **100** may be deemed unsuitable for deployment. Further, the tourniquet's embedded software version holds significance; an outmoded firmware may result in the system algorithm prioritizing an alternative tourniquet **100** with a more recent software iteration.

[0101] The primary intent of the tourniquet **100** is the swift application upon a patient's extremity, ensuring cessation of hemorrhagic flow and maintaining sufficient constriction (adequate to inhibit both arterial and venous circulation) pending medical interference. Under circumstances wherein a proximate tourniquet **100** presents with diminished energy levels, and alternatives with optimal energy reservoirs are further from the incident, it might still be deemed advantageous to engage the closer tourniquet, even if its energy suffices solely for initial constriction. Alternatively, if a manual bulb is known to be connected to the electronics unit, it may be deployed over a low battery tourniquet with just the electronic inflator.

[0102] Similarly, if the tourniquet **100** has undergone recent interactions, such as cuff replacements or pressurization assessments, the algorithm might dispatch such a tourniquet over others. Certain situations might necessitate the dispatch of multiple tourniquets **100**, taking into account the potential unavailability of certain responders or the varying transit times to the incident.

[0103] Illustratively, map **432** demonstrates potential tourniquets **100** for deployment. This high volume of tourniquets **100** may be common in urban areas. Tourniquets **100** in transit **437** and **438** are contained in an arrow symbol, while stationary units **434** or **435** are circumscribed within a circle. The investigative radius exhibited is relatively condensed, in alignment with the emergency's severity. In contrast, in less densely populated terrains, it might be advantageous to augment the search perimeter until a deployable tourniquet **100** is discerned.

[0104] As in this example, if a tourniquet **100** is on private property, such as in a home or office building, the Responder App **424** shows that it is a "Private TQ" **430**. Further, if it is owned by a Responder or volunteer responder, who register their willingness to serve as a Responder, volunteer responder, or both, for example both on and off duty, during initial device configuration, it is indicated by the title "Responder" **431**. If the user does not elect to be a Responder or volunteer responder, the tourniquet **100** will not appear on the map as an option to dispatch. If a public access tourniquet **100** is stored in an office building or gymnasium, the Responder App **424** would similarly show that the tourniquet **100** is a "Public TQ", which is similarly registered during initial configuration.

[0105] Tourniquets **100** can be dispatched by selecting "Request Dispatch" in the case of a

privately held TQ **430**, or “Dispatch” in the case of a responder held TQ **431**.

[0106] Once the request is submitted to the centralized server **500**, a notification **442** is sent which requests assistance, such as that in FIG. **4D**. A number of notifications can be delivered, for example by a cell phone **440** alert notification **442** shown in FIG. **4D**.

[0107] The mobile application in FIG. **4E** displays an option for controlling the tourniquet **100**. Prior to tourniquet application, the application displays the same content shown in FIGS. **4A-4D** depending on if the user is a standard user or a responder. During application, a notification is delivered that the tourniquet is active **444**. As aforementioned, the controls can be used on the mobile application in addition to the buttons on TQController **102**, such as the mute button **446**. The app displays pressure and allows for pressure adjustments **448**, shows time applied **450**, and location **452**. Pressure can be adjusted using the inflate and deflate feature. If a certain elapsed time period has passed, the app will present the option to slightly loosen and tighten the limb to provide reperfusion. The mobile application receives data from the centralized server **500** or directly from the tourniquet if connected via Bluetooth.

[0108] FIG. **5** shows the interaction between the centralized server **500** and a number of external systems, software, and stakeholders. The centralized server **500** receives and returns data and information from each connected section. Each section is a representation of many different potential connected devices, and the display of one of each section is not meant to be limiting since a broad number of potential connections can be made.

[0109] TQ Station **300** represents a TQ Station as shown in FIG. **3**. The station **300** collects information from each internal tourniquet (TQ **1.1**) **504** and (TQ **1.n**) **506**, as each tourniquet is connected to connector **321** in FIG. **3C**. Individual tourniquets outside of a station **100** can self-report to the centralized server **500**. Individual tourniquets **100** must be connected to a supporting device, for example a cell phone via Bluetooth, in order for each tourniquet's transmitter **206** in FIG. **2A** to report data to the centralized server **500**. As aforementioned, the TQ Controller **102** possesses the ability to provide detailed information about the tourniquet. This includes current whereabouts from GPS coordinates, a street address, building floor level, or room number/name. This data can be updated live from the locator module **204** or preset in FIG. **2A**. Additionally, the controller **102** stores and reports data, such as serial number, most recent cuff replacement, etc. Moreover, controller **102** maintains records of paired devices within the tourniquet network and indicates whether the device has recently been checked or interacted with by the owner. Lastly, it utilizes a comprehensive total device readiness field, which takes into account all the aforementioned factors to determine whether the device is prepared for immediate use or not.

[0110] The data aggregated in the centralized server **500** can be reported to various stakeholders through the User App **400** or Responder App **424**. In the case of an active emergency, the data may be reported to a PASP, Computer-Aided Dispatch (CAD), or 911 Dispatcher **514** and various connected responders **516**.

[0111] CAD software is commonly used by dispatchers to manage police cars, ambulances, and other emergency response vehicles. TQ data can be uploaded from the centralized server **500** to CAD software **514**. Alternatively, volunteer or professional responders can be deployed through the Responder App **424**. Once the dispatcher assigns units to the emergency location, the dispatcher can track them via the CAD system. The responders **516** gather the necessary tourniquets **100** for the rescue, navigate to the emergency, and use the tourniquet **100** to stop the bleeding. The patient is then transported to the hospital for further treatment.

[0112] Extremity hemorrhage can cause death in very little time; therefore it is desirable to dispatch the closest available tourniquet **100** to the scene of the emergency. The applications described herein allow for rapid dispatch of the closest available tourniquet **100** to reduce mortality. The centralized server **500** can use an algorithm or algorithms of varying complexity levels to determine if the tourniquet **100** is able to be deployed or not deployed based on different combinations of factors aforementioned, such as battery health, sync status, last interaction, total

device readiness, etc. The selection and dispatch of tourniquets **100** can involve a wide range of protocols, processes, and algorithms, which can vary significantly.

[0113] FIG. **6** demonstrates a few examples of how the tourniquet network can be activated for the purpose of deploying assistance to an emergency bleed. Network activation can begin after emergency bleeding occurs **602**. The centralized server **500** in FIG. **5** is either notified of a bleed **604**, which can occur for example through a call to EMS, or through the server receiving a request to stop a bleed through control **338** in FIG. **3D** which may also be integrated into either of the applications shown in FIGS. **4A-4C**. The centralized server then identifies nearby available tourniquets **606** shown in FIG. **4C** and delivers a request for assistance **608**. This request can be delivered through the onboard communicator **142** of the tourniquet in FIG. **2A**, through a notification from the TQ Station Monitor **306** and or communicator **329** in FIG. **3C**, through a phone notification in FIG. **4D**, through PASP or CAD software, or a 911 Dispatcher **514** in FIG. **5**.

[0114] During initial device configuration, users indicate their availability for emergency dispatch and their notification preferences. For example, in a school classroom setting, it may be undesirable to have audible notifications of nearby massive hemorrhages, unless they occur within the school as part of a mass casualty incident. In such cases, the tourniquet **100** can serve as a life-saving alert device by delivering an evacuation alert. The request for assistance is either accepted or denied by a user or multiple users. If denied, the request disappears, and depending on the availability of nearby tourniquets **100** or the frequency at which rejections are received more requests may be delivered. If the request is accepted, the tourniquet is deployed to stop bleeding **610**. It is sometimes desirable for the centralized server **500** to send multiple requests through multiple channels, as the rate by which users respond is unpredictable. In some embodiments, the tourniquet's transmitter **206** in FIG. **2A** includes a SIM card, which allows the 911 Dispatcher to talk to people present at the emergency and the patient through the speaker **142** and the user to talk back through the microphone **127** if included.

[0115] If the notification is accepted, the TQ App may open an external map site, such as Google Maps or Apple Maps, to provide navigation to the emergency coordinates. Alternatively, the User App can display an internal map **421** or **432** with directions.

[0116] Alternatively, an alert is sent to the centralized server **500** immediately upon tourniquet application **612**, then delivered to the centralized server **614**, causing a request for assistance to be delivered. In this case, it is less desirable to employ the User App **400** to have private tourniquets deployed to the emergency, as the bleeding has already stopped. The network then delivers a request for assistance **616** to EMS services. EMS is then either deployed **618**, or in the case of a military scenario or for example in an active shooter scenario, the casualty is displaced **620**. In that case it may be desirable to initiate a long-term care protocol until EMS arrives **622**, which may include periodic, automatic reperfusion to restore partial blood flow to a limb to prevent long term damage from tourniquet use.

[0117] Redundancies are built into the tourniquet **100** in case of device hardware or software failure. For example, if hardware fails, the system may send additional responders to the emergency to deploy another tourniquet **100**. An inflator bulb **133** such as in FIG. **1F** may be used to mechanically inflate the cuff **104**. In another case, the TQ Station **300** may receive the data that the tourniquet **100** has failed, and a different tourniquet **100** may be applied from the station **300**. This is possible due to the tourniquet's networking capabilities.

[0118] Whereas the present disclosure has been described in relation to the drawings attached hereto, other and further modifications, apart from those shown or suggested herein, may be made within the spirit and scope of this disclosure. Those skilled in the art will appreciate that the conception, upon which this disclosure is based, may readily be utilized as a basis for the designing of other structures, methods, and systems for carrying out the several purposes of the present disclosure. That is, the present disclosure is not limited in its application to the details of construction and to the arrangements of the components set forth in the following description or

illustrated in the drawings. The present disclosure is capable of other embodiments and of being practiced and carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting. The descriptions of the embodiments shown in the drawings should not be construed as limiting or defining the ordinary and plain meanings of the terms of the claims unless such is explicitly indicated. The claims should be regarded as including such equivalent constructions insofar as they do not depart from the spirit and scope of the present disclosure.

Claims

1. A method for stopping patient blood loss utilizing a networking-capable pneumatic emergency tourniquet that is capable of automatically applying and maintaining a patient-independent preset pressure around a patient's limb upon detection and activation of only two control switches or buttons, the networking-capable emergency tourniquet comprising electronics including a communication unit, a controller unit, an air pump, a microcontroller, one or more control switches or buttons, an inflatable member, and a power supply, the method comprising: after the networking-capable pneumatic tourniquet has been applied to the patient's limb, detecting, by the microcontroller, activation of a first control switch or button of the one or more control switches or buttons, upon which, without any intervening steps, activation of the networking-capable pneumatic tourniquet occurs; detecting activation of a second control button or switch; and upon the detection of the activation of the second control button or switch and without any intervening steps or receiving any input about the patient's physical characteristics, directing, by the microcontroller, the air pump to release air to pressurize the inflatable member to the preset pressure than can optionally increase if bleeding continues, the air pressurizing the inflatable member to stop blood loss.
2. The method of claim 1, further comprising: integrating a dual port valve comprising a top valve and a bottom valve into the inflatable member, the top valve of the dual port valve configured to interchangeably receive one of the controller or a mechanical inflation mechanism, the mechanical inflation mechanism configurable to be integrated with the microcontroller to maintain the preset pressure during use of the pneumatic tourniquet when the inflatable member is not attached to the controller.
3. The method of claim 1, further comprising programming the first control switch or button to toggle between idle and active mode rather than between on and off mode.
4. The method of claim 1, wherein the communication unit is configured to provide stepwise instructions.
5. The method of claim 4, wherein the stepwise instructions include one or more of pictorial and/or text instructions on a top of the controller unit facing a user or the patient.
6. The method of claim 5, further comprising complementing the pictorial instructions and/or the text instructions with audible instructions conveyed through the communication unit.
7. The method of claim 1, wherein after the networking-capable pneumatic tourniquet has been applied to the patient's limb, providing, by the communication unit, audible instructions to the patient or user to tighten the inflatable member above a bleeding area of the patient and to activate the second control switch or button.
8. The method of claim 1, further comprising: determining, by one or more sensors, that the networking-capable pneumatic tourniquet is powered on prior to being positioned above a bleeding site; and verifying correct placement of the tourniquet before initiating a tightening process.
9. A method for stopping patient blood loss utilizing a networking-capable pneumatic emergency tourniquet that is capable of automatically applying and maintaining a preset pressure around a patient's limb upon detection and activation of only two control switches or buttons, the networking-capable emergency tourniquet comprising electronics including a communication unit,

a controller unit, an air pump, a microcontroller, one or more control switches or buttons, an inflatable member, and a power supply, the method comprising: after the networking-capable pneumatic tourniquet has been applied to a patient's limb, detecting, by the microcontroller, activation of a first control switch or button of the one or more control switches or buttons, upon which, activation of the networking-capable pneumatic tourniquet occurs; detecting activation of a second control button or switch; and upon the detection of the activation of the second control button or switch, directing, by the microcontroller, the air pump to release air to pressurize the inflatable member to the preset pressure without receiving information regarding patient personalized pressure, the present pressure can optionally be increased if bleeding continues, the air pressurizing the inflatable member to stop blood loss.
